

CSE 2025

Remote Sensing C KEY

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This test consists of 50 multiple-choice questions worth two points each, and 6 multi-part free-response questions worth varying amounts of points. There are a total of 196 points.

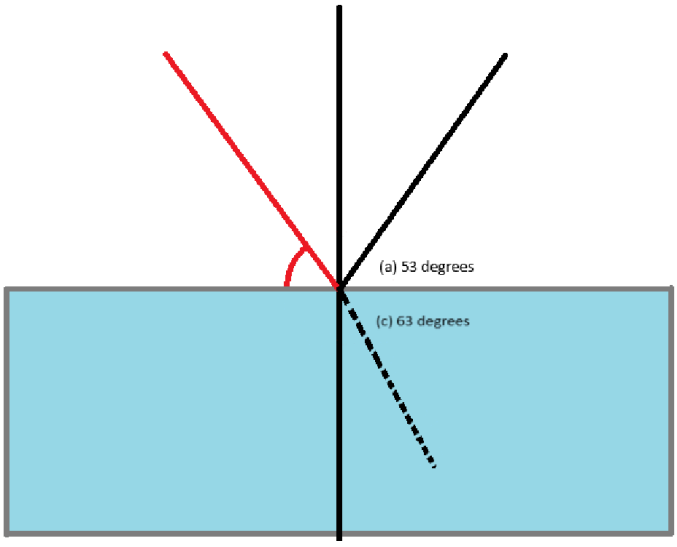
Multiple-choice; 2 points each

1. C	18. B	35. C
2. A	19. A	36. B
3. A	20. A	37. B
4. D	21. D	38. A
5. B	22. C	39. B
6. D	23. D	40. C
7. D	24. C	41. B
8. A	25. A	42. D
9. A	26. D	43. A
10. A	27. B	44. C
11. B	28. A	45. C
12. C	29. A	46. B
13. C	30. B	47. C
14. B	31. D	48. D
15. A	32. D	49. C
16. D	33. C	50. A
17. A	34. B	

Free-response; point values are indicated

51.

a. 2 points for drawing a solid line 53 degrees above the horizontal line as shown below.



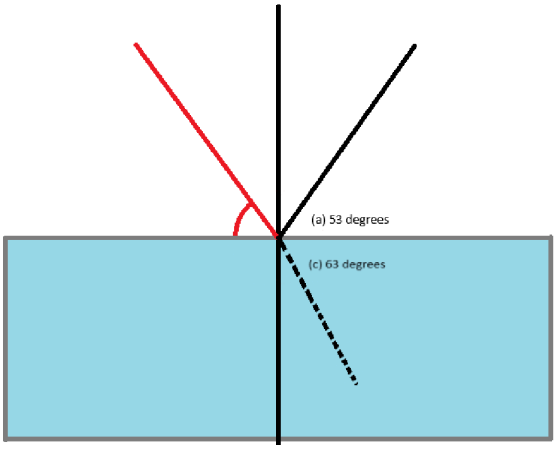
b. 1 point for using Snell's Law, $n_1 sin(\theta_1) = n_2 sin(\theta_2)$

1 point for plugging in $n_1 = 1.00,\ \theta_1 = 90 - 53 = 37\ degrees,\ and\ n_2 = 1.33$

2 points for obtaining $\theta_2 = 27\ degrees$

1 point for finding that the angle from the horizontal is 90 – 27 = 63 degrees.

c. 2 points for drawing a dashed line 63 degrees below the horizontal line as shown below.



d. 2 points for determining that the angle is 53 + 63 = 116 degrees (no work required).

e. 3 points for stating that the angle would increase (no explanation required).

f. 1 point for using the Beer-Lambert Law, $I = I_0 e^{-kd}$

2 points for plugging in $\frac{I}{I_0} = 0.01 \; and \; k = 0.6$

2 points for obtaining d = 7.68 meters

g. 2 points for identifying that sin(63) = depth / distance traveled

1 point for obtaining depth = 6.84 meters

52.

a. Up to 2 points for listing the following satellites (1 point per satellite): Terra, Aqua, Aura, Suomi NPP

b. 2 points for stating brown

3 points for a valid justification. Some examples include:

- Most common over the oceans, which is the source of sea spray
- Some areas (look at the western Pacific off the coast of Chile) match the shape of tropical cyclones, which cause sea spray

c. 2 points for stating light blue

3 points for a valid justification. Some examples include:

- Wildfires are common over California and Canada, which emit black carbon
- Natural and/or human-caused wildfires are common over Africa due to forests and savannas, which emit black carbon

d. 2 points for stating dark green

3 points for a valid justification. An example:

- Deserts, such as over western Africa and the Arabian Peninsula often have sand and dust storms which blow dust into the air

53.

a. 2 points for combustion of fossil fuels

b. 1 point for stating that the image represents an improvement of air quality

2 points for mentioning that NO_2,which is responsible for many health-related issues (such as respiratory problems), is decreasing in concentration overall

c. 3 points for mentioning reduced burning of grasslands by wildfires and/or controlled burns

d. 2 points for mentioning that consumption of fossil fuels is increasing in Africa

2 points for mentioning that increased fossil fuel use should lead to a higher concentration of NO_2

54.

a. 1 point for mentioning specific marine organisms (phytoplankton/algae)
   1 point for stating that photosynthesis traps CO_2 from the atmosphere
   1 point for stating that marine organisms die, sink, and carry their carbon to the ocean floor
b. 1 point for stating that CO_2 directly dissolves into surface water from the air
   2 points for mentioning oceanic circulation, which can carry surface waters to the deep ocean.
c. 1 point for stating higher latitudes
   1 point for stating that higher latitudes are cooler
   1 point for stating that CO_2 is more soluble in cold water
d. 1 point for stating that as the atmosphere warms, more CO_2 will be released from solution by the ocean
   2 points for stating that this causes a warming effect (due to the greenhouse effect)
   2 points for stating that this is a positive feedback loop

55.

a. 2 points for Gravity Recovery and Climate Experiment
b. 1 point for gravitational anomalies
c. 1 point for each of the following:
   - The first satellite is pulled slightly ahead of the second, causing the distance between them to increase
   - After the first satellite passes the anomaly, it slows down
   - The distance between the satellites decreases
   - The second satellite passes the anomaly, causing it to speed up, causing the distance between them to decrease
   - After the second satellite passes the anomaly, it slows down
   - The distance between the satellites increases

56.

a. 3 points for $R(1 - a)$
b. 3 points for $Ra + \sigma T^4$
c. 2 point for $R(1 - a) = \sigma T^4$
   2 points for $\sqrt[4]{\frac{R(1-a)}{\sigma}}$
d. 1 point for $R(1 - a)$
   1 point for σT_A^4
   2 points for $R(1 - a) = \sigma T_A^4$
e. 1 point for σT_S^4
   1 point for $R(1 - a) + \sigma T_A^4$
   2 points for $\sigma T_S^4 = R(1 - a) + \sigma T_A^4$
f. 1 point for stating that $T_S > T_A$
   2 points for substituting the equation from part (d) into the equation from part (e)
   1 point for obtaining $\sigma T_S^4 = 2\sigma T_A^4$